UCI 2123: Systems Engineering with AWS

Capstone Project

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1.1 Problem Definition

The Travel Management Website is a web-based application that enables users to create, manage, and organize trip-related data within a centralized system. It provides functionality for storing structured travel information—such as itineraries, tickets, and uploaded documents—while offering calendar-based visualization for scheduling and planning. The system is designed with a scalable architecture and secure data handling to support efficient access, organization, and future integration capabilities.

1.2 Requirements Documentation

1. Problem statement and business justification
    - Planning trips often involves juggling multiple tools—emails for confirmations, notes for itineraries, and separate apps for flights, hotels, and activities, while enhancing time efficiency in planning. This fragmented approach leads to disorganization, missed details, and inefficient planning.

      The proposed **Travel Management Website** will provide a centralized platform where users can create trips, store travel-related documents (emails, tickets, PDFs), and organize itineraries in one place.

2. Target users and user stories (minimum 8 user stories)
   (see image below for Monday.com work management place)
   https://chantraphom555s-team.monday.com/boards/18404809671
    - Target users: frequent travelers, business travelers, family vacation

    1) As a user, I want to create a new trip so I can organize my travel plans.
    2) As a user, I want to add flight information so I can track departure and arrival details.
    3) As a user, I want to store tickets (flight, bus, train) so I can access them in one place.
    4) As a user, I want to upload files (PDFs, confirmations) so I don't lose important documents.
    5) As a user, I want to add links (e.g., booking websites) so I can revisit them easily.
    6) As a user, I want to create a daily itinerary so I can plan activities for each day.
    7) As a user, I want to edit or delete trip details so I can keep my plans updated.
    8) As a user, I want to store email confirmations so I can reference booking details.

- Functional requirements (minimum 10)
  1. The system shall allow users to create, view, edit, and delete trips.
  2. The system shall allow users to input transportation details (flight, road trip, etc.).
  3. The system shall allow users to upload and store files (PDFs, attachments).
  4. The system shall allow users to save and manage external links.
  5. The system shall allow users to create and manage itineraries by day.
  6. The system shall display trip information in a calendar view.
  7. The system shall allow users to add multiple ticket types (flight, event, transportation).
  8. The system shall provide a dashboard displaying all user trips.
  9. The system shall allow users to update and delete stored information.

- Non-functional requirements (performance, security, scalability)

⚡ Performance

- The application shall load within 2–3 seconds under normal conditions.
- The system shall handle multiple user requests efficiently without noticeable delay.
- File uploads and retrieval should occur within acceptable response times.

🔐 Security

- The system shall require user authentication (login/signup).
- User data shall be securely stored and protected from unauthorized access.
- File uploads shall be validated to prevent malicious content.

📈 Scalability

- The system shall support an increasing number of users and stored trips.
- The architecture shall allow for future integration with third-party APIs.
- The database shall efficiently handle growing volumes of data.

Out-of-scope items

- Expense tracking or budgeting tools
- Currency conversion tools
- Syncing with external calendars (Google Calendar, Outlook)

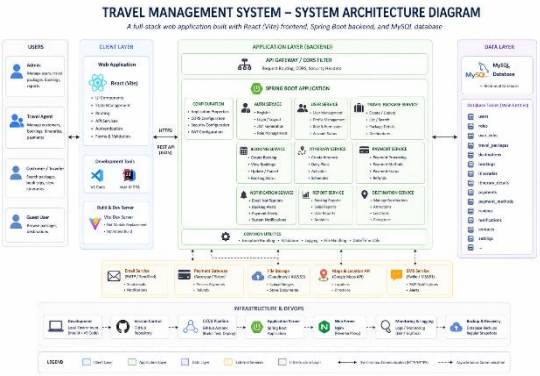
1.3 Architecture Design

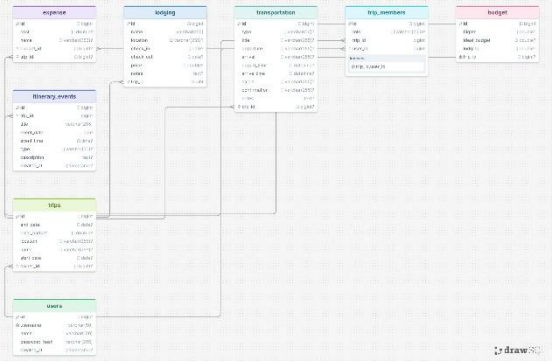
System Arhchitect Diagram

Entity-Relationship Diagram

API List

Component Design





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